

Rishindra Davuluri

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EDUCATION

The University of Texas at Austin *M.S. Computer Science, Integrated B.S./M.S.* May 2028
Texas CSB Honors Program *B.S. Computer Science Honors, B.B.A. Business Honors* May 2027

GPA: 3.81 | Coursework: Machine Learning I & II, Algorithms, Operating Systems, Computer Architecture, Linear Algebra

EXPERIENCE

Loopfour (YC W23) - *AI Engineer Intern*; Mountain View, CA Jun 2026 - Present

- Building AI copilot artifact system that maps natural language to **30+ agent tools** and executable workflows
- Engineering tool-calling engine on **Vercel AI SDK** with Claude & Zod-typed tools and a multi-step streaming agent loop

Moody College of Communication - *NLP Researcher*; Austin, TX Nov 2025 - Present

- Building and deploying an **OpenAI**-powered platform (React, FastAPI, AWS) for behavioral study
- Classified **300,000** complaints using anchor-based semantic similarity with **sentence embeddings** and cosine similarity
- Engineered **BERTopic** and regex feature-engineering pipeline to analyze demographic disparities in financial complaints

UT Austin: Learning Directed Operating System - *Systems Researcher*; Austin, TX Oct 2025 - Feb 2026

- Designed lightweight **C protocol** for learned OS components to share runtime signals and synchronize decision-making
- Benchmarked protocol on **CloudLab** bare-metal instances using learned prefetcher and allocator agents

Capital One - *Software Engineer Intern*; McLean, VA Jun 2025 - Aug 2025

- Designed and implemented **Python/AWS Lambda** omni-channel cache library backed by **Snowflake** data
- Integrated cache into Vue.js and Vuex webpage, improving workflow completion by **40%** for **80K** monthly fraud concerns
- Built internal security-awareness simulation with Next.js and Twilio to deliver training emails and measure click engagement

Fidelity Investments - *Software Engineer Intern*; Westlake, TX Jun 2024 - Aug 2024

- Enhanced batch job responsible for **100M+** updates to central database with SQL & MyBatis
- Improved processing by **20%** by developing **multi-threaded Java pipeline** with cloud integration

PROJECTS

Low-Latency Matching Engine | *C++, UDP, Linux*

- Building **C++20** limit-order-book engine with price-time priority, cancels, partial fills, and deterministic tests
- Profiling CPU affinity and cache-aware order-book layout to reduce scheduler migrations and **improve hot-path latency**

Ace: Proactive AI Study Agent | *Python, Go, Gemini API, Angular, gRPC, Google Cloud, Neo4j*

- Architected AI tutoring platform (**Go, Python, gRPC**) deployed on GCP with **Neo4j** and BigQuery for telemetry
- Engineered **Gemini RAG pipeline** with LLM-as-a-judge guardrails, quiz generation, and customizable lessons

Operating System & Graphics | *C++, Python, Assembly, QEMU, Bash, Linux, Multi-threading*

- Developed core C++ and Assembly kernel: interrupt handling with **$O(1)$ helper threads**; system calls; semaphores
- Engineered virtual memory, **Ext2** file system with **third-level indirection**, and **VBE** graphics subsystem

LEADERSHIP EXPERIENCE AND ACTIVITIES

Texas Product Engineering Organization - *Engineering and Product Fellow*; Austin, TX Sep 2024 - Present

- Developing **Python web crawlers**, PHP plugins, and websites with React, REST, and Vercel for nonprofits
- Built **Locus** with TypeScript, Node.js, and **Firebase** for **sub-100ms** location and status updates

ADDITIONAL INFORMATION

Technical Languages: Python, C++, Java, C, Go, TypeScript, SQL

Cloud, Data, & Infrastructure: AWS, GCP, Docker, GitHub Actions, Snowflake, Terraform, Neo4j, Salesforce

Frameworks, Environments, & Tools: PyTorch, NumPy, Pandas, scikit-learn, FastAPI, React, Node.js, Linux, gdb

Awards & Certifications: Eagle Scout, Azure AI Fundamentals

Languages: Fluent in English and Telugu; Intermediate in Spanish and Hindi

Interests: Texas Infinities (Gender-Inclusive Spirit & Service), Singing, Guitar, Camping, Ping Pong, Drawing, Soccer